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Peer Review File

Manipulating charge transfer dynamics and stabilizing lead bromide octahedra for efficient blue perovskite light-emitting diodes

Corresponding Author: Professor Xuyong Yang

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This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
In this manuscript, Zhang et al. reported a short-chain ligand IDMP for preparing reduced-dimensional pure bromide perovskites. The IDMP-treated LEDs showed blue emission with a peak EQE of 16.3% at 474 nm with an operating half-life of 392 min; sky-blue emission with a peak EQE of 25.4% at 487 nm and an operating half-lifetime of 845 min. The topic holds significant interest for the field, however, there are some issues to be addressed before further consideration of this manuscript in Nature Communications.

1. I wonder whether IDMP would serve as A cation in the perovskite lattice or not. Please provide more structural characteristics.
2. The authors claimed that IDMP could anchor with the perovskite through double-ends, but I am worried about the steric hindrance. Please provide both experimental and theoretical evidence.
3. The authors measured time-dependent PL images of pristine and IDMP-treated perovskite films under an external electric field, it is interesting but confusing.
  Please provide experimental details including the preparation of samples.
  How about the AMP-treated films?
  How about the time-dependent PL images of pristine and IDMP-treated perovskite films under an external electric field in LEDs?
4. The authors claimed that the EL performance was still hindered by inferior charge transport/ injection capability and deep-level defect-induced nonradiative exciton loss. I wonder whether the enhanced device performance in this work is related to improvements in these specific aspects, and suggest the authors to provide more evidence of suppression of deep-level defects in the material.
5. The characterization of charge transport properties is essential. The authors have provided the mobility data (6.53×10^{-5} $cm^2 V^{-1} s^{-1}$ and 9.41×10^{-5} $cm^2 V^{-1} s^{-1}$), please compare the obtained mobility values with those of other state-of-the-art blue-emitting perovskite materials.
6. There is a shift in the EL peak in Fig. 4b, why?
7. The authors should provide complete J–V curves corresponding to the devices in Fig. 4d. The maximum EQE of the control device occurs at a low current density, what is the corresponding luminance? These data are essential for a comprehensive evaluation of device performance.
8. In Supplementary Table 4, please provide a complete comparison with the reported blue-emitting perovskite LEDs, including mixed halide perovskite counterparts (for example, Sci. Adv. 2024, 10, eado5645, EQE of 24.3% at 475 nm, and 26.4% at 480 nm).

Reviewer #2

(Remarks to the Author)
The authors have reported highly-efficient blue LEDs by using a di-functional phosphonic acid as a ligand for Br-based perovskites. In general, the data is well-presented, and many characterization studies and computational studies were

provided. The paper should be publishable in Nature Communications upon some revision.

1) I personally prefer the pure-blue (474nm) LED results, which the authors have put into the SI. This is a much more useful wavelength, and is probably a lot more difficult to get good efficiencies compared to sky-blue LEDs. It does not seem to do the data much justice to have them in the SI. I recommend moving it into the main text.

2) Can the authors comment on why they are getting effectively 100% IQE performance for their LEDs, but their PL quantum yield is notably lower at ~70%?

3) Can the authors provide more experimental information on how their LEDs are measured? This is especially important when a 25% EQE (which is approximately 100% IQE) results is reported.

4) The authors did not show the J-V results below 3V. Is the full scan performed or is the data omitted below 3V for some reasons?

5) The authors show a 400mm2 device in their photos, but I believe the characterization data belongs to a 4mm2 device. I think this is misleading, and should either be clearly explained in the caption and main text, or should be presented as separate figures (also with clear specification on the size of the device for the device data).

6) This is a common problem in LED papers, but is there any way that the authors can get their high efficiencies validated? Unlike solar cells where there are independent accredited institutions and commercial labs for efficiency validation, many LED results reporting top efficiencies are not validated. It would be good if the authors can provide some form of validation for this.

Reviewer #3

(Remarks to the Author)
In this manuscript, the authors report highly efficient and stable blue perovskite LEDs that employ a short-chain iminodi(methylphosphonic acid) (IDMP) ligand with two terminal phosphonic acid groups to modulate charge-transfer dynamics, reduce defect density, and stabilize the lead bromide octahedra in layered perovskites. As a result, the champion device achieves a peak external quantum efficiency (EQE) of 25.4% and an operational lifetime exceeding 800 minutes, representing record performance among previously reported pure-Br perovskite-based blue LEDs. This work presents a simple and effective strategy for achieving high-performance blue PeLEDs and is likely to attract broad interest in the field of perovskite LEDs. However, there are still some questions need to be addressed before the reviewer can recommend its publication in Nature Communications.

1.In the wok, the authors didn’t discuss how the ratio of PBA/MOPA affects the emission performance, which should be provided for better highlighting the roles of the IDMP molecules. Besides, the details about the deposition of perovskite films should be provides in the Experiment Section, for example, any antisolvent is applied or not?
2.Since the IDMP molecules are demonstrated to be very effective in improving the performance and stability of the perovskite films, have the authors consider to apply it as an interlayer in the PeLEDs ? If so, what’s the effect? If not, please explain the reasons.
3.In this work, the authors demonstrated the strategy in the sky-blue and pure-blue PeLEDs, how about the effect of the IDMP molecules in fabricating deep-blue perovskite films and devices? It still works or not?
4.Could the authors give some comparisons between this work with the previously reported blue LEDs which based on mixed halide perovskites? This can help the community to evaluate the influence of this work better.
5.In Fig. 4h, the test conditions (including environment, and temperature,) should be explicitly stated in the main text to support the evaluation and comparison.
6.There are some format issues need to be addressed. For instance, the calculation of charge density, the author should provide the coordinate axes and mark the elements represented by each color.

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
The revised manuscript is now suitable for Nature Communications.

Reviewer #2

(Remarks to the Author)
The revisions are satisfactory and I recommend acceptance of the manuscript for publication.

Reviewer #3

(Remarks to the Author)

The authors have fully revised the manuscript according to the reviewers's comments, and I recommend that the article be accepted as it is.

Reply to Reviewer #1

In this manuscript, Zhang et al. reported a short-chain ligand IDMP for preparing reduced-dimensional pure bromide perovskites. The IDMP-treated LEDs showed blue emission with a peak EQE of 16.3% at 474 nm with an operating half-life of 392 min; sky-blue emission with a peak EQE of 25.4% at 487 nm and an operating half-lifetime of 845 min. **The topic holds significant interest for the field**, however, there are some issues to be addressed before further consideration of this manuscript in Nature Communications.

We sincerely appreciate the reviewer's very helpful comments on our manuscript.

1. I wonder whether IDMP would serve as A cation in the perovskite lattice or not. Please provide more structural characteristics.

Reply:

The electroneutral IDMP molecules (**Fig. R1a**) cannot serve as an A-site cation in the quasi-2D perovskite lattices with a structural formula of L_nABX_3. For our case, the A-site cations refer to the monovalent Cs cations, and the L represents larger organic intercalating cations of PBA^+ and $MOPA^+$, which both have single-terminal amino group. During the growth of perovskite, the larger L-cations can partially replace the small A-cations and inhibits the 3D growth of $[BX_6]^{4-}$ along out-of-plane directions and thus, cutting the 3D perovskites into quasi-2D slices (*Nature, 2024, 631, 73-79; Nat. Commun. 2023, **14**, 397; Adv. Mater. 2022, 34, 2205217*), as schematically shown in **Fig. R1b**. However, the IDMP molecules are organic acid with two-terminated phosphonic groups, which although can interact with ($[PbBr_6]^{4-}$) octahedra *via* synergetic covalent bonds (Pb-O-P), but cannot function as the spacer cations.

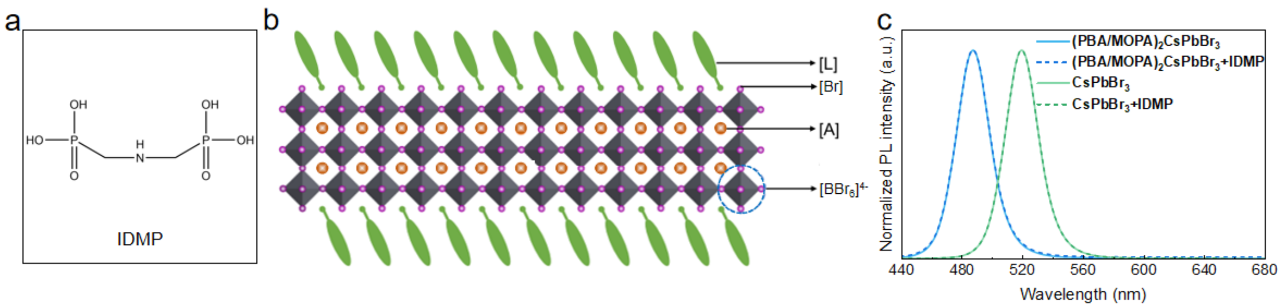

Figure R1. (a) The molecule formula of IDMP. (b) A schematic illustration of typical quasi-2D perovskite (n=3) and the chemical structures of large organic ligands. (c) The PL spectra for films of $(PBA/MOPA)_2CsPbBr_3$ and $CsPbBr_3$ perovskite films without and with IDMP treatment.

2. The authors claimed that IDMP could anchor with the perovskite through double-ends, but I am worried about the steric hindrance. Please provide both experimental and theoretical evidence.

Reply:

If the interactions between IDMP and octahedra affect the steric hindrance of the perovskite system, it is plausible to expected the shift of the PL peaks when increasing the amount of IDMP molecules (*Adv. Mater. 2024, 36, 2308670, ACS Nano 2025, 19, 12182-12193,*).

As requested, in experimentally, we have increased the amount of IDMP molecules in the precursor solutions to check whether their steric hindrance plays role or not. As show in **Fig. R2a**, when varying the employed amounts of IDMP molecules in the precursor solutions, it only affects the PL intensity and there in no change in the position of the PL peak, indicating the IDMP molecules cannot affect the phase compositions of the perovskites. This is also consistent with the UV-vis absorption spectra results (**Fig. R2b**), as we have discussed in our manuscript "*Notably, the optical bandgap (E_g) for both AMP- and IDMP-treated samples remained unchanged (Supplementary Fig. 4). These results imply that the AMP and IDMP ligands hardly affected the quantum confinement effect but improved the dielectric confinement in quasi-2D pure-Br perovskites*." The slight decrease of PL intensity for sample with 3 mg /mL IDMP is due to that the limited solubility of IDMP in dimethyl sulfoxide (DMSO) affects the crystallization of perovskite, and therefore, we utilized the 2 mg/mL for all target samples in our work.

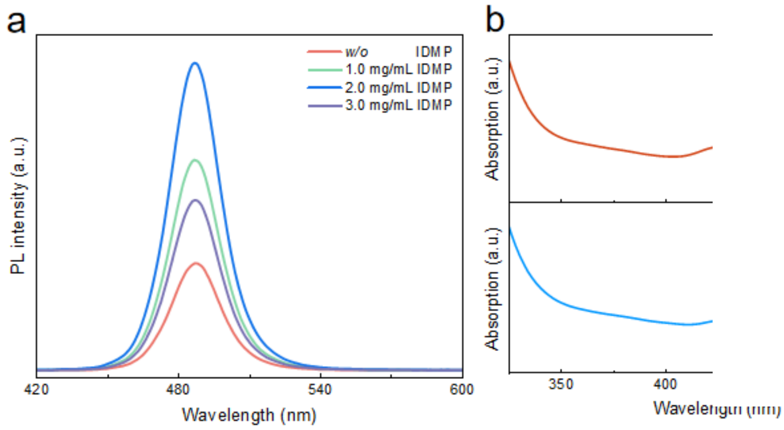

Figure R2. (a) The PL spectra for pristine and different concentrations of IDMP-treated perovskite films. (b, Figure S4 in SI) Steady-state absorption spectra for pristine and IDMP-treated perovskite

3. The authors measured time-dependent PL images of pristine and IDMP-treated perovskite films under an external electric field, it is interesting but confusing.

We have carefully revised the related descriptions in our manuscript according to the following questions raised by the reviewer.

(1) Please provide experimental details including the preparation of samples.

Reply:

For this test, the sample has a configuration of glass/perovskite/Ag. The perovskite film is spin-coated onto the glass substrate and followed by thermal annealing treatment (60 °C for 5 minutes), on which two parallel Ag electrodes with a spacing of 200 μm was deposited by thermal evaporation. And we utilized a custom-built setup which contains a commercial optical microscope with integrated lighting source (Nikon ECLIPSE L200N), and a source meter unit (KUAIQU SPS-E305). The test is performed at ambient environment. A constant bias of 6.0 V was imposed across the samples, and the microscope images were collected at a 60-second interval to monitor the evolution of the samples under the illumination of internal lighting source (with a central wavelength of ~ 460 nm).

The related information has been added in the *Characterization Section* (Page 15, Lines 5-11) and highlighted in yellow in our revised manuscript. The details are as follows:

"*For time-dependent microscopic images, the samples with a configuration of glass/perovskite/Ag were fabricated, where two parallel Ag electrodes with a spacing of 200 μm was deposited on the perovskite films. The custom-built measurement setup contains a commercial optical microscope with integrated lighting source (Nikon ECLIPSE L200N) and a source meter unit (KUAIQU SPS-E305). A constant bias of 6.0 V was imposed on the samples and the microscope images were collected at a 60-second interval at the ambient environment*."

(2) How about the AMP-treated films?

Reply:

As requested, we have measured the corresponding images for AMP-treated films (**Fig. R3**). It was observed that there is severe degradation of the AMP-treated perovskite film occurred at 180 s due to ionic migration.

The related discussion has been updated in our revised manuscript (Page 10, Lines 15-16):

"The pristine film and AMP-treated film showed severe PL degradation and spatial phase segregation after 120 s and 180 s, respectively."

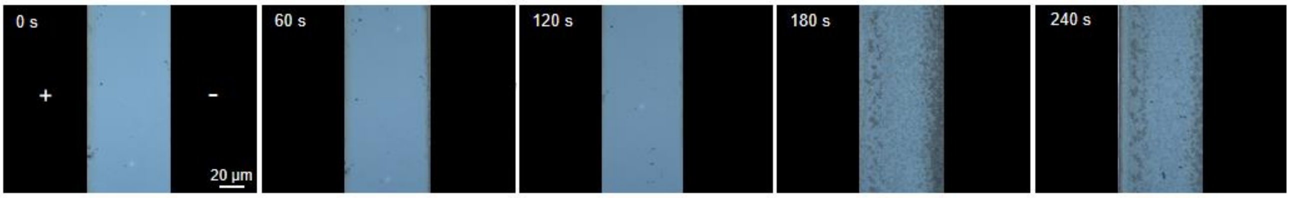

Figure R3 (Supplementary Figure 19 in revised Supplementary Information) Time-dependent microscopic images of AMP-treated perovskite films under an external electric field (~6×10^4 V m^{-1}). Scale bars, 20 μm. The bilateral black areas refer to Ag electrodes and the external electric field is marked by the “+” and “-” symbols.

(3) How about the time-dependent PL images of pristine and IDMP-treated perovskite films under an external electric field in LEDs?

<u>Reply</u>:

Thanks for your question. However, since the LED is vertically stacked structure and one side of the device is opaque metal Al electrode. Under the current set-up system, the bottom light will be totally reflected by the Al electrode and it cannot irradiate the perovskite layer, which means that we cannot observe and collected the PL image of the films in LEDs directly.

4. The authors claimed that the EL performance was still hindered by inferior charge transport/ injection capability and deep-level defect-induced nonradiative exciton loss. I wonder whether the enhanced device performance in this work is related to improvements in these specific aspects, and suggest the authors to provide more evidence of suppression of deep-level defects in the material.

<u>Reply</u>:

The enhanced device performance in this work is indeed related to the improvement of charge transport/injection capability and the suppression of deep-level defects.

As we have discussed in our manuscript, the IDMP molecules actually play roles in both the crystallization stage of perovskite films and the operation of device under electrical bias. Specifically, the strong interactions between IDMP molecules and lead bromide octahedra can effectively passivate shallow- and deep-level defects, for which the corresponding defect density was reduced to 1.0×10^{17} from 4.6×10^{17} cm^{-3} as derived from the SCLC results (Fig. S16). This leads to the remarkable enhancement of the PLQY from 21% to 70% (Fig. 1 in the main text). The IDMP molecules also enhanced the conductivity of the perovskite films (Fig. 3d in the main text), which is desired for efficient charge transport/injection capability in LEDs. When the film was integrated into the LED, the reduced C_p value reveals the suppressed interfacial charge accumulation induced by inefficient charge injection/transport (Fig. 3e in the main text). Besides, the improved charge balance also benefits from the enhanced charge transport/injection capability (**Fig. R4**).

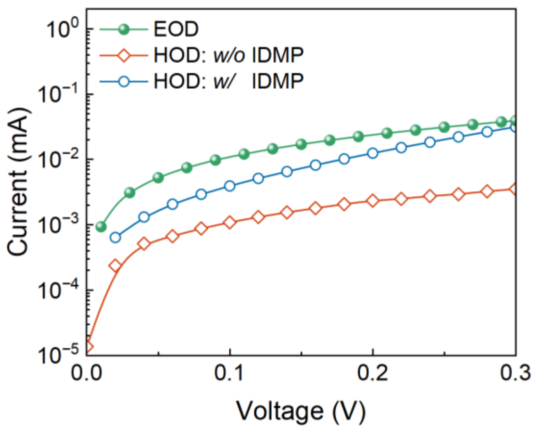

Figure R4. Current-voltage curves of the electron-only device (EOD) and hole-only devices (HODs) based on pristine and IDMP-treated perovskite films.

5. The characterization of charge transport properties is essential. The authors have provided the mobility data (6.53×10^{-5} cm^2 V^{-1} s^{-1} and 9.41×10^{-5} cm^2 V^{-1} s^{-1}), please compare the obtained mobility values with those of other state-of-the-art blue-emitting perovskite materials.

Reply:

As requested, we have summarized the reported values of carrier mobility for state-of-the-art blue-emitting perovskite materials, including perovskite quantum dots, pure bromine and mixed halogen perovskite films (**Table R1**). The order of magnitude of carrier mobility varies in the range 10^{-10}~10^{-4}, and the measured data in our results are within this range and is relatively high among the reported values. Thanks.

Table R1. Summary of the charge transport mobility for state-of-the-art blue-emitting perovskite materials.

Perovskite	EL peak [nm]	EQE [%]	Charge transport mobility [cm^2 V^{-1} s^{-1}]	Reference
$CsPbBr_3$ QDs	480	18	3.5×10^{-3}	1
$(FA/Cs)Pb(Br_x/Cl_{1-x})_3$	502	21.5	8×10^{-4}	2
$DADOCs_{n-1}Pb_nBr_{3n+1}$	493	16.55	4.56×10^{-10}	3
$PEA_2(Cs)_{n-1}Pb_nBr_{3n+1}$	467	4.42	1.11×10^{-4}	4
$(MOPA/PBA)_2Cs_{n-1}Pb_nBr_{3n+1}$	**487**	**25.4**	**9.41×10^{-5}**	**our work**

1. Jiang, Y. et al. Synthesis-on-substrate of quantum dot solids. *Nature* **612**, 679-684 (2022).
2. Zheng, S. et al. Stable sky-blue perovskite LEDs achieved by efficient sub-bandgap emission with 24.5% PCE. *Adv.*

Funct. Mater. e21079 (2025).
3. Zhao, H., Xuan, T., Bai, W., Dong, H., Dong, G., Xie, R-J. Fluorine polymer-modified Dion-Jacobson perovskites enabling sky-blue perovskite light-emitting diodes with an efficiency of 16.55%. *Laser Photonics Rev.* **17**, 2300430 (2023).
4. Zhang, F. et al. High-performance blue perovskite light-emitting diodes enabled by synergistic effect of additives. Nano Lett. 24, 1268-1276 (2024).

6. There is a shift in the EL peak in Fig. 4b, why?

Reply:

We have carefully compared the data and drew a reference line (**Fig. R6**), and it shows that there is almost no shift occurred in the EL spectra. In case any possible visual error, we have added the reference line and updated the figure in our revised manuscript (Page 11). Thanks.

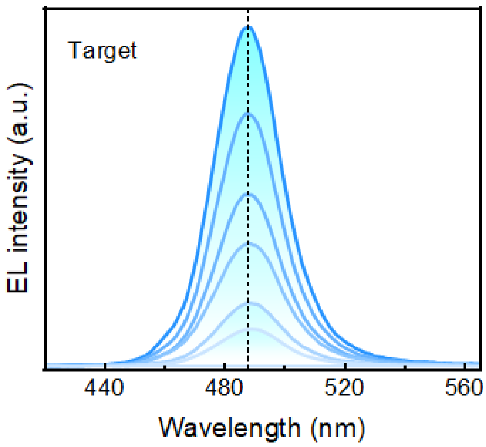

Figure R6. EL spectra for the target device with varying voltages.

7. The authors should provide complete *J*-*V* curves corresponding to the devices in Fig. 4d. The maximum EQE of the control device occurs at a low current density, what is the corresponding luminance? These data are essential for a comprehensive evaluation of device performance.

Reply:

As requested, we have updated the complete *J*-*V* curves (**Fig. R7a**) into the **Fig. 4d** of our revised manuscript. The corresponding luminance at the maximum EQE was 89 cd m^{-2} and 253 cd m^{-2} for control and target devices, respectively (**Fig. R7b**).

The higher current density for target device at low voltages indicates more efficient charge injection, considering the improved film qualityfor the IDMP-treated perovskite film (as discussed for Supplementary Fig. 13 in our Supporting Information). And the efficiency roll-off at high brightness in our target device is obviously alleviated, highlighting the advantage of our proposed strategy. These data indeed help provide more direct and comprehensive proofs to evaluate the performance and reinforce our conclusions. Thanks.

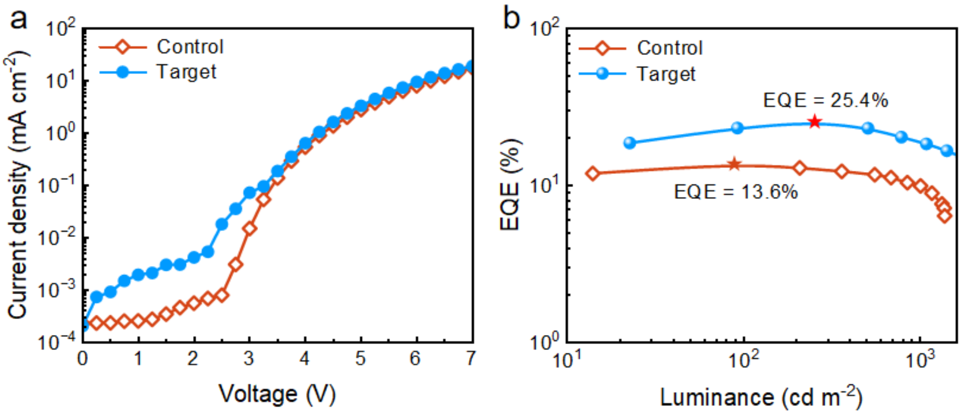

Figure R7. (a) *J-V* and (b) *EQE-L* curves for control and target PeLEDs.

8. In Supplementary Table 4, please provide a complete comparison with the reported blue-emitting perovskite LEDs, including mixed halide perovskite counterparts (for example, Sci. Adv. 2024, 10, eado5645, EQE of 24.3% at 475 nm, and 26.4% at 480 nm).

Reply:

As requested, we have revised and updated **Supplementary Table 5** (**Table R2**) in our revised Supporting Information. The as-mentioned literature (*Sci. Adv. 2024, 10, eado5645)* has been cited as ref. 9. The newly added literatures are highlighted in yellow for your convenient read.

Table R2. Summary of performance parameters of representative blue LEDs based on pure-Br and mixed halide perovskites reported in literatures.

	Perovskite	λ_{EL} (nm)	Luminance (cd m^{-2})	EQE (%)	Stability (T_{50})	Ref
deep-blue	$CsPbBr_3$ nanoplatelets (NPLs)	455	1511	4.15	50 min @ 100 cd m^{-2}	1
	$CsPbBr_3$ NPLs	460	591	1.60	N/A	2
	$CsPbBr_3$ NPLs	462	691	1.77	20 min @ 100 cd m^{-2}	3
	$CsPbBr_3$ NPLs	464	4.6	0.70	N/A	4
	$CsPbBr_3$ NPLs	465	227	5.4	860 s @ 100 cd m^{-2}	5
	$PEA_2(Cs_{0.6}EA_{0.4}PbBr_3)_2PbBr_4$	466	N/A	3.64	81 s @ 30 cd m^{-2}	6
	$(CHDA_{1-x}CHMA_x)(Cs_{1-m}FA_m)_{n-1}Pb_nBr_{3n+1}$	466	121	9.22	125 s @ 11 cd m^{-2}	7
	$(p\text{-}FPEA)_2Cs_{1.5}Pb_{2.5}Br_{8.5}$	469	2428	10.4	57 min @ 100 cd m^{-2}	8
	$CsPb(Br_x/Cl_{1-x})_3$ NCs	460	255	12.0	~75 s @ 1 mA cm^{-2}	9
	$CsPb(Br/Cl)_3$	461	1978	5.68	4.1 min @ 100 cd m^{-2}	10
	$PEA_2(Cs/FA)_{n-1}Pb_n(Br/Cl)_{3n+1}$	467	448.81	4.42	78 s @ 100 cd m^{-2}	11
	$CsPb(Br/Cl)_3$	468	282	16.3	N/A	12
	$p\text{-}F\text{-}PEABr_2Cs_{n-1}Pb_n(Br/Cl)_{3n+1}$	468	~100	15	253 s @ 1 mA cm^{-2}	13
	$CsPb(Br/Cl)_3$ QDs	469	31407	20.3	T_{80} 4.8 h @ 100 cd m^{-2}	14
pure	$PBA_2(Cs_{1-m}FA_mPbBr_3)_{n-1}PbBr_4$	474	~ 200	4	N/A	15

-	$PEA_2(CsPbBr_3)_{n-1}PbBr_4$	478	512	5.52	80 s @ 99 cd m^{-2}	16
blue	$CsPb(Br_x/Cl_{1-x})_3$ NCs	470	635	21.3	~80 s @ 1 mA cm^{-2}	9
	$CsPbCl_{3-x}Br_x$ QDs	470	1176	24.9	77 min @ 100 cd m^{-2}	17
	o-F-$PEA_2(GA/Cs)_{n-1}Pb_n(Br_x/Cl_{1-x})_{3n+1}$	472	1297	9.32	15 min @ 100 cd m^{-2}	18
	(p-F-PEA/PBA)$_2$(Li/GA/Cs)Pb(Br/Cl)$_3$	472	267.5	15.1	T_{70}: 16 min @ 100 cd m^{-2}	19
	$(PEA/PPA/EDAD)_2Cs_{n-1}Pb_n(Br_x/Cl_{1-x})_{3n+1}$	476	943	16.61	N/A	20
	$(FA/Rb/Cs)Pb(Br_x/Cl_{1-x})_3$	478	8136	10.6	N/A	21
	$(MOPA/PBA)_2Cs_{n-1}Pb_nBr_{3n+1}$	**474**	**826**	**16.3**	**392 min @ 100 cd m^{-2}**	**Our work**
	$CsPbBr_3$ quantum dots (QDs)	480	~ 2000	18	2 h @ 100 cd m^{-2}	22
	$PBA_2(FA/Cs)_{n-1}Pb_nBr_{3n+1}$	483	~ 700	9.5	250 s @ 100 cd m^{-2}	23
	$(PA/PEA)_2CsPbBr_3$	486	1061	16.2	600 s @ 0.1 mA cm^{-2}	24
	$PEA_2(Cs_{1-x}EA_xPbBr_3)_2PbBr_4$	488	2191	12.1	1 h @ 100 cd m^{-2}	25
	$(BDA_{1-x}PEA_{2x})(CsPbBr_3)_{y-1}PbBr_4$	489	716	13.7	154 s @ 100 cd m^{-2}	26
	$DADOCs_{n-1}Pb_nBr_{3n+1}$	493	3028	16.55	52 s @ 100 cd m^{-2}	27
	$BEA_2Cs_{n-1}Pb_nBr_{3n+1}$	493	2540	16.98	730 s @ 0.5 mA cm^{-2}	28
sky-	$CsPb(Br_x/Cl_{1-x})_3$ NCs	480	1601	26.4	~100 s @ 1 mA cm^{-2}	9
blue	$(PEA/PPA/EDAD)_2Cs_{n-1}Pb_n(Br_x/Cl_{1-x})_{3n+1}$	485	4384	24.03	160 min @ 100 cd m^{-2}	20
	$(FA/Rb/Cs)Pb(Br_x/Cl_{1-x})_3$	487	18748	13.6	N/A	21
	$(FA/Cs/Rb)(Pb/Sr)(Br/Cl)_3$	487	5700	23.3	42 min @ 50 cd m^{-2}	29
	(PEA/p-F-PEA)$_2(K/Cs)_{n-1}Pb_n(Br_x/Cl_{1-x})_{3n+1}$	488	3321	9.56	175 min @ 100 cd m^{-2}	30
	$PEA_2(FA/Cs)Pb(Br_x/Cl_{1-x})_3$	490	~1500	25.87	720 s @ 1 mA cm^{-2}	31
	$(FA/Cs)Pb(Br_x/Cl_{1-x})_3$	502	7449	21.5	318.8 min @ 1000 cd m^{-2}	32
	$(MOPA/PBA)_2Cs_{n-1}Pb_nBr_{3n+1}$	**487**	**2545**	**25.4**	**845 min @ 100 cd m^{-2}**	**Our work**

References

1. Liu, H. et al. Organic semiconducting ligands passivated $CsPbBr_3$ nanoplatelets for blue light-emitting diodes. *ACS Energy Lett.* **8**, 4259-4266 (2023).
2. Huang, Q. et al. Enhancing crystal integrity and structural rigidity of $CsPbBr_3$ nanoplatelets to achieve a narrow color-saturated blue emission. *Light Sci. Appl.* **13**, 111 (2024).
3. Liu, H. et al. Efficient and stable blue light emitting diodes based on $CsPbBr_3$ Nanoplatelets with surface passivation by a multifunctional organic sulfate. *Adv. Energy Mater.* **13**, 2201605 (2023).
4. Oshita, Naoaki. et al. Pure-blue emitting face-down-oriented perovskite nanoplatelets for light-emitting diodes. *ACS Appl. Nano Mater.* **8**, 8588-8594 (2025).
5. Wang, L. et al. Efficient, color-stable, pure-blue light-emitting diodes based on aromatic ligand-engineered perovskite nanoplatelets. *Nano Lett.* **24**, 7004-7011 (2024).
6. Dong, F. et al. Partially ionized amino acid salt enabling spatially vertical n-phase distribution of quasi-2D perovskite for efficient deep-blue light-emitting diodes. *Adv. Optical Mater.* **12**, 2401247 (2024).

7. Hu, J. et al. Monoammonium modified dion-jacobson quasi-2D perovskite for high efficiency pure-blue light emitting diodes. *Small* **20**, 2402786 (2024).
8. Wei, K. et al. Managing edge states in reduced-dimensional perovskites for highly efficient deep-blue LEDs. *Adv. Mater.* **37**, 2412041 (2024).
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10. Lee, S. et al. Brightening deep-blue perovskite light-emitting diodes: A path to Rec. 2020. *Sci. Adv.* **10**, eadn8465 (2024).
11. Zheng, X. et al. Self-assembled mono-hole-injecting layers towards efficient deep-blue perovskite light-emitting diodes. *Appl. Surf. Sci.* **655**, 159643 (2024).
12. Cao, L. et al. In situ interface reaction enables efficient deep-blue perovskite light-emitting diodes. *Angew. Chem. Int. Ed.* **64**, e202513617 (2025).
13. Zhang, K. et al. Nondestructive halide exchange via S_N2-like mechanism for efficient blue perovskite light-emitting diodes. *Nat. Commun.* **15**, 10621 (2024).
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26. Peng, X. et al. Suppressed energy transfer loss of Dion-Jacobson perovskite enabled by DMSO vapor treatment for efficient sky-blue light-emitting diodes. *ACS Energy Lett.* **8**, 339 (2023).

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29. Chen, Y. et al. All-site alloyed perovskite for efficient and bright blue light-emitting diodes. *Nat. Commun.* **16**, 3254 (2025).
30. Li, C. et al. Hexamethylene diisocyanate-modulated mixed-halide perovskite enables efficient and stable blue light-emitting diodes. *ACS Appl. Mater. Interfaces* **17**, 55045-55053 (2025).
31. Yu, Y. et al. Efficient blue perovskite LEDs via bottom-up charge manipulation for solution-processed active-matrix displays. *Adv. Mater.* **37**, 2503234 (2025).
32. Zheng, S. et al. Stable sky-blue perovskite LEDs achieved by efficient sub-bandgap emission with 24.5% PCE. *Adv. Funct. Mater.* e21079 (2025).

Reply to Reviewer #2

The authors have reported highly-efficient blue LEDs by using a di-functional phosphonic acid as a ligand for Br-based perovskites. In general, **the data is well-presented, and many characterization studies and computational studies were provided**. The paper should be publishable in Nature Communications upon some revision.

We appreciate the reviewer for the positive and helpful comments on our manuscript.

1. I personally prefer the pure-blue (474 nm) LED results, which the authors have put into the SI. This is a much more useful wavelength, and is probably a lot more difficult to get good efficiencies compared to sky-blue LEDs. It does not seem to do the data much justice to have them in the SI. I recommend moving it into the main text.

Reply:

Thanks for your valuable suggestions. We have now moved the data for pure-blue devices into the main text and updated Fig. 4 **(Fig. R8)** in our revised our manuscript.

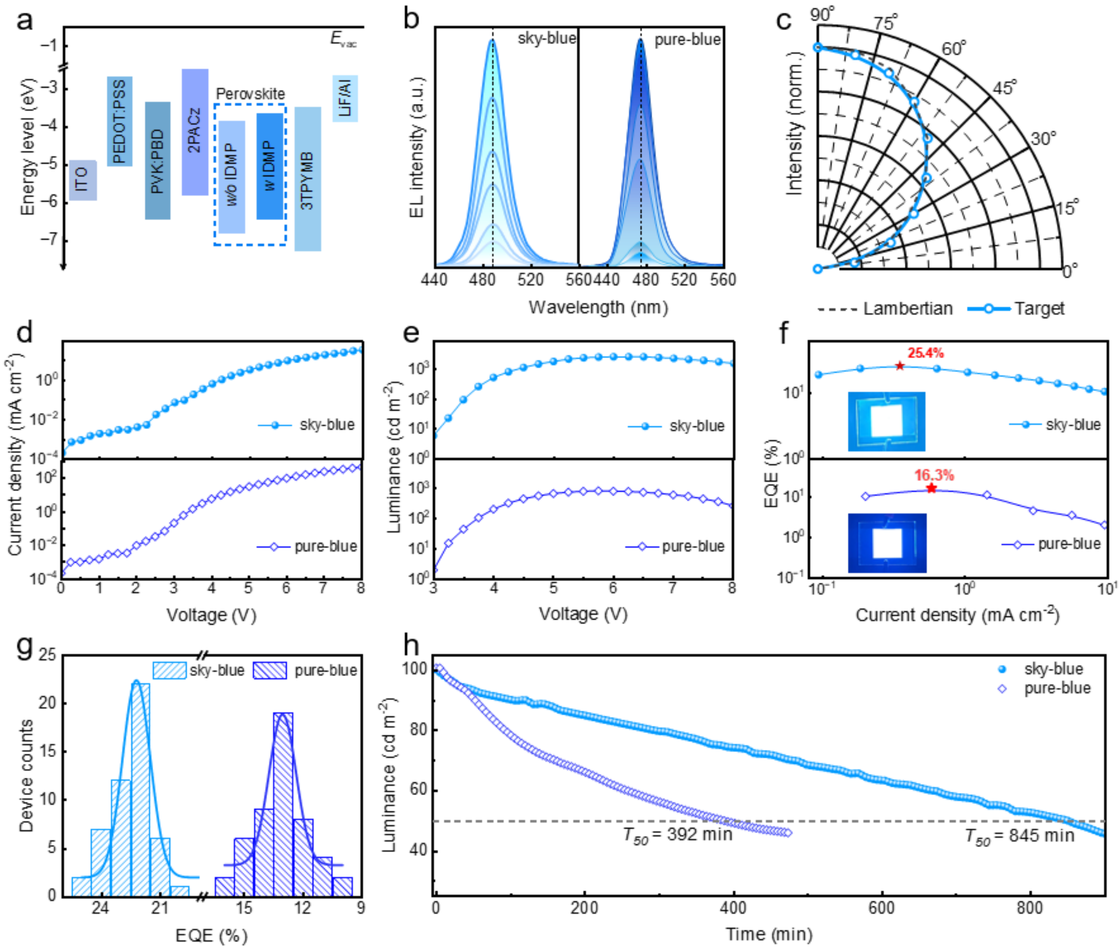

Figure R8 (Figure 4 in revised manuscript) | Device performance of the sky-blue and pure-blue PeLEDs. (a)The flat band energy level diagram of the PeLEDs. (b) EL spectra for devices under varying voltages. (c) Angle-dependent EL intensity for the sky-blue device. (d) *J-V*, (e) *L-V*, (f) *EQE-J* curves. The active area of the device for performance testing is 2 × 2 mm^2. The inset photographs in **f** dispaly operating target device (20 × 20 mm^2) under 4 V. (a)Histogram statistics of peak EQEs for sky-blue and pure-blue devices, and (h) the comparison for their corresponding operational lifetime.

2. Can the authors comment on why they are getting effectively 100% IQE performance for their LEDs, but their PL quantum yield is notably lower at ~70%?

Reply:

We now better explain on how the PLQY can affect the IQE in perovskite LEDs. Indeed, the ~100% IQE should correspond to a high PLQY of close to 100% in electroluminescent devices. It is worth point out that in our case the PLQY of 70% for the perovskite film was measured at a relatively low excitation power (~10 mW, FluoroMax$^+$, HORIBA), while for the perovskite emitters it will increase as the increase in excitation power (**Fig. R9a**) due to the enhanced fraction of bimolecular recombination and trap-filling behavior, thus may lead to a higher EQE of PeLEDs with the increase in carriers charge injection. This has been widely reported in previous literatures (*Nature 2022, 612, 679; Nature 2024, 630, 631-635; Nature 2024, 631, 73; Nat. Nanotechnol. 2025, 20, 507; Nat. Commun. 2025, 16, 3254*).

Moreover, we also conducted the power-dependent PL measurements on the IDMP-treated perovskite film. As shown in **Fig. R9b**, the PL intensity also increases with elevating excitation power, which suggests that a higher PLQY value can be obtained upon increasing the power intensity.

[FIGURE REDACTED]

Figure R9. (a) Excitation intensity-dependent PLQEs for control and dual-additive perovskites (*Nature 2024, 630, 631-635*). (b) Power-dependent PL intensity of perovskite films.

3. Can the authors provide more experimental information on how their LEDs are measured? This is especially important when a 25% EQE (which is approximately 100% IQE) results is reported.

Reply:

As requested, we have added more details and updated the related description in the "*Characterization Section*" of our revised manuscript (Page 14, Lines 24-28, and Page 15, Lines 1-2).

The details are as follow:

"The performance of the devices including EL spectra, J-V, L-V, I-V, and EQE-J characteristics were measured on a test system containing a QE Pro650 spectrometer, a source meter (Keithley 2400), and an integrating sphere (50 mm in diameter; Shenzhen Pynect). The as-fabricated device was placed in the sample chamber of the integrating sphere and get connected with power probe. The driving voltages was imposed from 0 V (with a step of 0.25 V) until the device gets burnt down, the data spot was collected with a fix integration time of 100 ms. All the above tests are performed on device without encapsulation and carried out in a nitrogen-filled glovebox at room temperature." Thanks.

4. The authors did not show the *J-V* results below 3 V. Is the full scan performed or is the data omitted below 3 V for some reasons?

Reply:

As shown in **Fig. R10**, the *J-V* curve below 3 V for the devices were plotted. And we have also updated the corresponding *J-V* curves in **Fig. 4d** (**Fig. R8d** in this letter) in our revised manuscript.

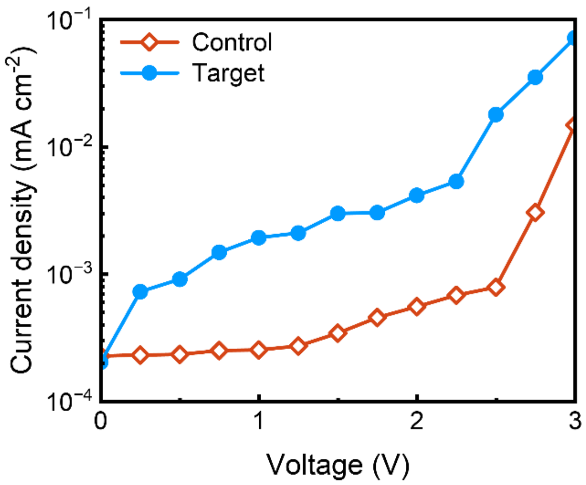

Figure R10. *J-V* curves below 3 V for control and target PeLEDs.

5. The authors show a 400 mm^2 device in their photos, but I believe the characterization data belongs to a 4 mm^2 device. I think this is misleading, and should either be clearly explained in the caption and main text, or should be presented as separate figures (also with clear specification on the size of the device for the device data).

Reply:

Thanks for your valuable suggestion. To avoid confusion, we have now clearly marked the corresponding device areas in the caption, as shown below.

"*...The active area of the device for performance testing is 2 × 2 mm^2. The insets photographs in f display operating target devices (20 × 20 mm^2) under a voltage of 4 V...*" *(Page 11, Line 4-6)*.

6. This is a common problem in LED papers, but is there any way that the authors can get their high efficiencies validated? Unlike solar cells where there are independent accredited institutions and commercial labs for efficiency validation, many LED results reporting top efficiencies are not validated. It would be good if the authors can provide some form of validation for this.

Reply:

As suggested, we have conducted the validation experiments by fabricating and testing our target device in another Lab (**Fig. R11** and **Table R3**), which is in Jilin University and under the charge of Prof. Xiaoyu Zhang (*orcid.org/0000-0001-7528-4405*). The very slight reduction of the EQE in their lab is within the reasonable range, which provides a cross-validation for the high device performance.

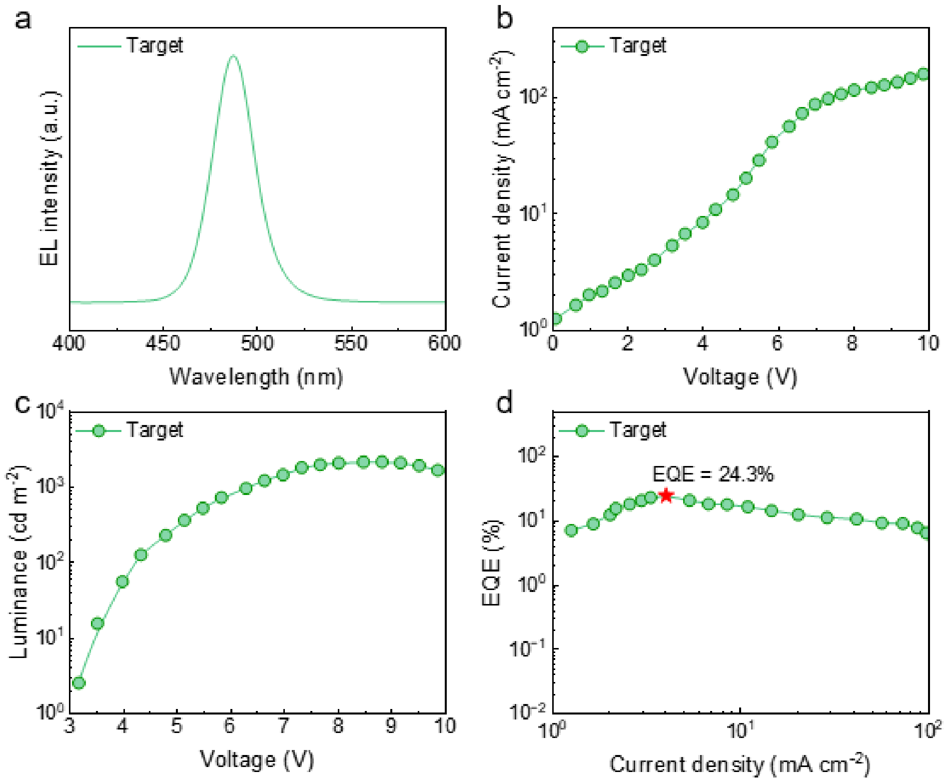

Figure R11. Device performance for the as-fabricated PeLED @ JLU. (a) EL spectrum, (b) *J-V*, (c) *L-V*, and (d) *EQE-J* curves.

Table R3. Comparison of the EL performance for the target devices fabricated in different labs.

Target PeLED	λ_{EL} (nm)	V_{on} (V)	Luminance (cd m^{-2})	EQE (%)
in our lab	487	3.0	2459	25.4
in JLU lab	487	3.0	2141	24.3

Reply to Reviewer #3

In this manuscript, the authors report highly efficient and stable blue perovskite LEDs that employ a short-chain iminodi(methylphosphonic acid) (IDMP) ligand with two terminal phosphonic acid groups to modulate charge-transfer dynamics, reduce defect density, and stabilize the lead bromide octahedra in layered perovskites. As a result, the champion device achieves a peak external quantum efficiency (EQE) of 25.4% and an operational lifetime exceeding 800 minutes, representing record performance among previously reported pure-Br perovskite-based blue LEDs. **This work presents a simple and effective strategy for achieving high-performance blue PeLEDs** and is likely to attract broad interest in the field of perovskite LEDs. However, there are still some questions need to be addressed before the reviewer can recommend its publication in Nature Communications.

We thank the reviewer for the positive comments on our work.

1. In the wok, the authors didn't discuss how the ratio of PBA/MOPA affects the emission performance, which should be provided for better highlighting the roles of the IDMP molecules. Besides, the details about the deposition of perovskite films should be provides in the Experiment Section, for example, any antisolvent is applied or not?

Reply:

As suggested, we have added the related discussions and more details about the deposition of perovskite films in our revised manuscript.

As shown in **Fig. R12**, by varying the ratio of PBA/MOPA, the phase compositions and the corresponding blue emission wavelengths can be well adjusted. Typically, in our work, the molar ratios of PBA/MOPA we used are 0.08:0.04 and 0.12:0.04 for preparing films with emission at 487 nm and 474 nm, respectively.

We indeed used ethyl acetate as the antisolvent to induce the rapid formation of perovskite nuclei during in the fabrication of perovskite films.

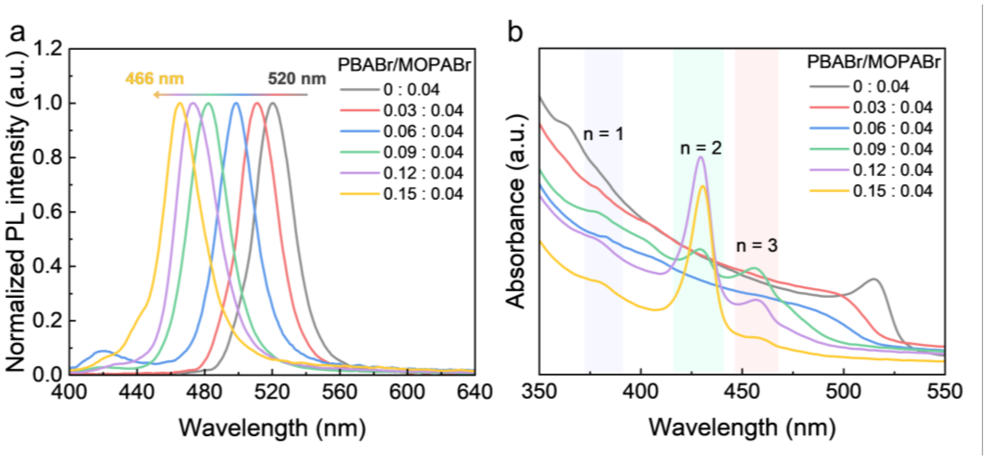

Figure R12 (Supplementary Figure 22 in revised Supplementary Information). (a) PL spectra and (b)

UV-vis absorption of perovskite films based on PBA/MOPA with different molar ratios.

The revised contents are listed as follows:

"…We obtained both sky-blue (λ_{em} = 487 nm) and pure-blue (λ_{em} =474 nm) devices by adjusting the proportions of spacer cations of PBA^+ and $MOPA^+$ accordingly (*Supplementary Fig. 22*)…. "

"*Perovskite was spin-coated at 3000 rpm for 60 seconds during which 240 μL of ethyl acetate was rapidly dispensed as an anti-solvent onto the film at 25~30 second window, and baked at 60 °C for 5 minutes.*" (Page 14, Lines 17-19).

2. Since the IDMP molecules are demonstrated to be very effective in improving the performance and stability of the perovskite films, have the authors consider to apply it as an interlayer in the PeLEDs? If so, what's the effect? If not, please explain the reasons.

Reply:

As suggested, we have explored its application as an interfacial layer in our device (**Fig. R13**). The overall EL performance of the LED containing the interlayer of IDMP shows nearly no enhancement as compared to those for control device. In our work, we mainly focused on the multiple roles of IDMP on improving the comprehensive properties of the perovskite film, if the IDMP only serve as interlayer, its potential will be limited.

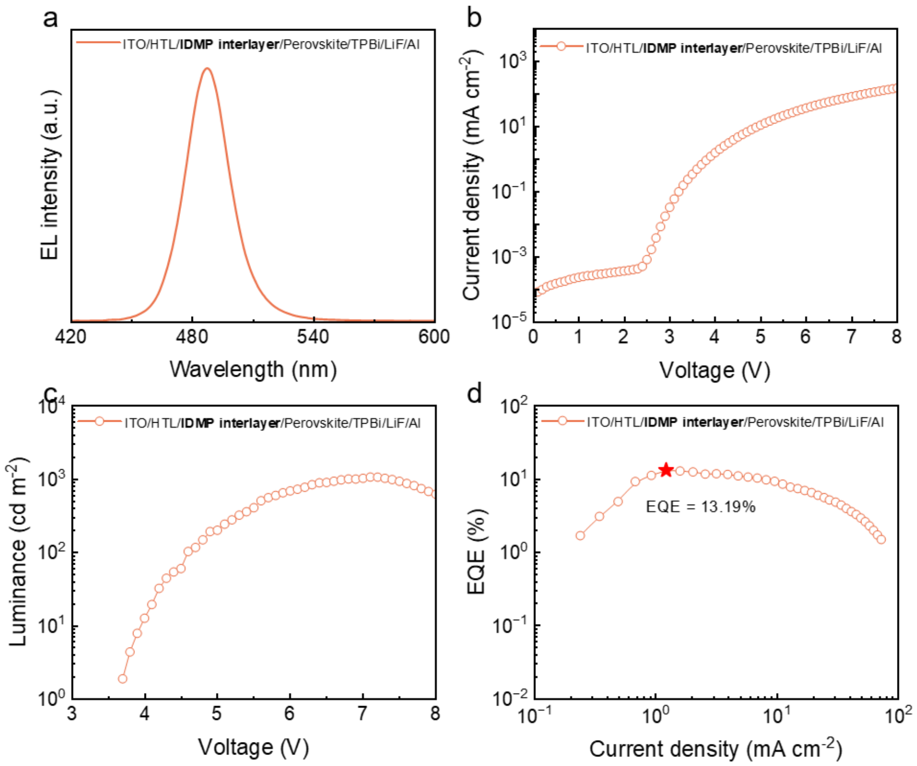

Figure R13. EL performance for PeLEDs with a IDMP interlayer. (a) EL spectrum, (b) *J-V*, (c) *L-V*, (d) *EQE-J* curves.

3. In this work, the authors demonstrated the strategy in the sky-blue and pure-blue PeLEDs, how about the effect of the IDMP molecules in fabricating deep-blue perovskite films and devices? It still works or not?

Reply:

As requested, we also employed the IDMP in molecules to fabricate deep-blue perovskite films and devices and it still works, as shown in (**Fig. R14** and **Table R4**). The brightness of the device has increased by 3.7 times compared to the control device (from 38 cd m^{-2} to 141 cd m^{-2}), and the EQE increased by more than twice from 1.8% to 3.0%.

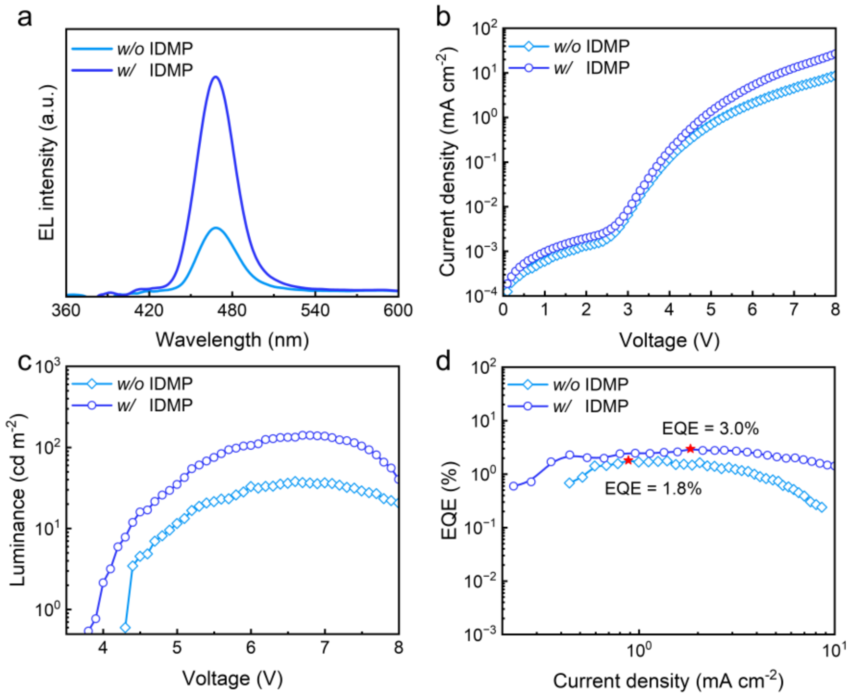

Figure R14 (Supplementary Figure 26 in revised Supporting Information). EL performance for deep-blue PeLEDs based on pristine and IDMP-treated perovskite films. (a) EL spectra peaking at 468 nm, (b) *J-V*, (c) *L-V*, (d) *EQE-J* curves.

Table R4 (Table S4 in revised Supporting Information). Summary of performance for deep-blue LEDs based on pristine and IDMP-treated perovskite films.

PeLED	EL peak (nm)	Luminance (cd m^{-2})	EQE (%)
w/o IDMP	468	38	1.8
w IDMP	468	141	3.0

4. Could the authors give some comparisons between this work with the previously reported blue LEDs which based on mixed halide perovskites? This can help the community to evaluate the influence of this work better.

Reply:

As requested, we have summarized the performance of the mixed halide perovskite LEDs and updated into **Supplementary Table 5** of our revised Supporting Information. For your convenience, the table is listed as below, in which the added data are highlighted in yellow. Thanks.

Table R5. Summary of performance parameters of representative blue LEDs based on pure-Br and mixed halide perovskites reported in literatures.

	Perovskite	λ_{EL} (nm)	Luminance (cd m^{-2})	EQE (%)	Stability (T_{50})	Ref
deep-blue	$CsPbBr_3$ nanoplatelets (NPLs)	455	1511	4.15	50 min @ 100 cd m^{-2}	1
	$CsPbBr_3$ NPLs	460	591	1.60	N/A	2
	$CsPbBr_3$ NPLs	462	691	1.77	20 min @ 100 cd m^{-2}	3
	$CsPbBr_3$ NPLs	464	4.6	0.70	N/A	4
	$CsPbBr_3$ NPLs	465	227	5.4	860 s @ 100 cd m^{-2}	5
	$PEA_2(Cs_{0.6}EA_{0.4}PbBr_3)_2PbBr_4$	466	N/A	3.64	81 s @ 30 cd m^{-2}	6
	$(CHDA_{1-x}CHMA_x)$ $(Cs_{1-m}FA_m)_{n-1}$ Pb_nBr_{3n+1}	466	121	9.22	125 s @ 11 cd m^{-2}	7
	$(p\text{-}FPEA)_2Cs_{1.5}Pb_{2.5}Br_{8.5}$	469	2428	10.4	57 min @ 100 cd m^{-2}	8
	$CsPb(Br_x/Cl_{1-x})_3$ NCs	460	255	12.0	~75 s @ 1 mA cm^{-2}	9
	$CsPb(Br/Cl)_3$	461	1978	5.68	4.1 min @ 100 cd m^{-2}	10
	$PEA_2(Cs/FA)_{n-1}Pb_n(Br/Cl)_{3n+1}$	467	448.81	4.42	78 s @ 100 cd m^{-2}	11
	$CsPb(Br/Cl)_3$	468	282	16.3	N/A	12
	$p\text{-}F\text{-}PEABr_2Cs_{n-1}Pb_n(Br/Cl)_{3n+1}$	468	~100	15	253 s @ 1 mA cm^{-2}	13
	$CsPb(Br/Cl)_3$ QDs	469	31407	20.3	T_{80} 4.8 h @ 100 cd m^{-2}	14
pure-blue	$PBA_2(Cs_{1-m}FA_mPbBr_3)_{n-1}PbBr_4$	474	~ 200	4	N/A	15
	$PEA_2(CsPbBr_3)_{n-1}PbBr_4$	478	512	5.52	80 s @ 99 cd m^{-2}	16
	$CsPb(Br_x/Cl_{1-x})_3$ NCs	470	635	21.3	~80 s @ 1 mA cm^{-2}	9
	$CsPbCl_{3-x}Br_x$ QDs	470	1176	24.9	77 min @ 100 cd m^{-2}	17
	$o\text{-}F\text{-}PEA_2(GA/Cs)_{n-1}Pb_n(Br_x/Cl_{1-x})_{3n+1}$	472	1297	9.32	15 min @ 100 cd m^{-2}	18
	$(p\text{-}F\text{-}PEA/PBA)_2(Li/GA/Cs)Pb(Br/Cl)_3$	472	267.5	15.1	T_{70}: 16 min @ 100 cd m^{-2}	19
	$(PEA/PPA/EDAD)_2Cs_{n-1}Pb_n(Br_x/Cl_{1-x})_{3n+1}$	476	943	16.61	N/A	20
	$(FA/Rb/Cs)Pb(Br_x/Cl_{1-x})_3$	478	8136	10.6	N/A	21
	$(MOPA/PBA)_2Cs_{n-1}Pb_nBr_{3n+1}$	**474**	**826**	**16.3**	**392 min @ 100 cd m^{-2}**	**Our work**
sky-blue	$CsPbBr_3$ quantum dots (QDs)	480	~ 2000	18	2 h @ 100 cd m^{-2}	22
	$PBA_2(FA/Cs)_{n-1}Pb_nBr_{3n+1}$	483	~ 700	9.5	250 s @ 100 cd m^{-2}	23
	$(PA/PEA)_2CsPbBr_3$	486	1061	16.2	600 s @ 0.1 mA cm^{-2}	24
	$PEA_2(Cs_{1-x}EA_xPbBr_3)_2PbBr_4$	488	2191	12.1	1 h @ 100 cd m^{-2}	25
	$(BDA_{1-x}PEA_{2x})(CsPbBr_3)_{y-1}$ $PbBr_4$	489	716	13.7	154 s @ 100 cd m^{-2}	26
	$DADOCs_{n-1}Pb_nBr_{3n+1}$	493	3028	16.55	52 s @ 100 cd m^{-2}	27
	$BEA_2Cs_{n-1}Pb_nBr_{3n+1}$	493	2540	16.98	730 s @ 0.5 mA cm^{-2}	28

$CsPb(Br_x/Cl_{1-x})_3$ NCs	480	1601	26.4	~100 s @ 1 mA cm^{-2}	9
$(PEA/PPA/EDAD)_2Cs_{n-1}Pb_n(Br_x/Cl_{1-x})_{3n+1}$	485	4384	24.03	160 min @ 100 cd m^{-2}	20
$(FA/Rb/Cs)Pb(Br_x/Cl_{1-x})_3$	487	18748	13.6	N/A	21
$(FA/Cs/Rb)(Pb/Sr)(Br/Cl)_3$	487	5700	23.3	42 min @ 50 cd m^{-2}	29
$(PEA/p\text{-}F\text{-}PEA)_2(K/Cs)_{n-1}Pb_n(Br_x/Cl_{1-x})_{3n+1}$	488	3321	9.56	175 min @ 100 cd m^{-2}	30
$PEA_2(FA/Cs)Pb(Br_x/Cl_{1-x})_3$	490	~1500	25.87	720 s @ 1 mA cm^{-2}	31
$(FA/Cs)Pb(Br_x/Cl_{1-x})_3$	502	7449	21.5	318.8 min @ 1000 cd m^{-2}	32
$(MOPA/PBA)_2Cs_{n-1}Pb_nBr_{3n+1}$	**487**	**2545**	**25.4**	**845 min @ 100 cd m^{-2}**	**Our work**

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16. Li, H. et al. Control of n-phase distribution in quasi two-dimensional perovskite for efficient blue light-emitting

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18. Yu, B. et al. Multifunctional biguanide additive stabilizes chloride-rich quasi 2D perovskites for efficient and stable pure-blue LEDs. *ACS Appl. Mater. Interfaces* **17**, 45927–45936 (2025).
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20. Chu, Z. et al. Blue perovskite light-emitting diodes using multifunctional small molecule dopants. *Adv. Mater.* **37**, 2409718 (2025).
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23. Liu, Y. et al. Efficient blue light-emitting diodes based on quantum-confined bromide perovskite nanostructures. *Nat. Photonics* **13**, 760-764 (2019).
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25. Chu, Z. et al. Large cation ethylammonium incorporated perovskite for efficient and spectra stable blue light-emitting diodes. *Nat. Commun.* **11**, 4165 (2020).
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29. Chen, Y. et al. All-site alloyed perovskite for efficient and bright blue light-emitting diodes. *Nat. Commun.* **16**, 3254 (2025).
30. Li, C. et al. Hexamethylene diisocyanate-modulated mixed-halide perovskite enables efficient and stable blue light-emitting diodes. *ACS Appl. Mater. Interfaces* **17**, 55045-55053 (2025).
31. Yu, Y. et al. Efficient blue perovskite LEDs via bottom-up charge manipulation for solution-processed active-matrix displays. *Adv. Mater.* **37**, 2503234 (2025).
32. Zheng, S. et al. Stable sky-blue perovskite LEDs achieved by efficient sub-bandgap emission with 24.5% PCE. *Adv. Funct. Mater.* e21079 (2025).

5. In Fig. 4h, the test conditions (including environment, and temperature,) should be explicitly stated in the main text to support the evaluation and comparison.

Reply:

All the relevant tests are performed on the device without further encapsulation at room temperature (25 °C) in a glove box filled with nitrogen. We have already added the relevant details in the main text.

The revised content is listed as follows (Page 12, Lines 16-17):

"… compare the operational stability of the PeLEDs, which were tested in a glove box (25 °C) filled with nitrogen without encapsulation…."

6. There are some format issues need to be addressed. For instance, the calculation of charge density, the author should provide the coordinate axes and mark the elements represented by each color.

Reply: